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Programme and Course Outline
Year 11 Mathematics : Methods
Stages 1 & 2

The program you have been issued references all objectives in the Australian Curriculum. The syllabus may be easily downloaded from the Schools Curriculum and Standards Authority (SCASA) website at (http://www.scsa.wa.edu.au/).

UNIT Name of course: Mathematics Methods

Rationale

Mathematics is the study of order, relation and pattern. From its origins in counting and measuring, it has evolved in highly sophisticated and elegant ways to become the language now used to describe much of the modern world. Statistics are concerned with collecting, analysing, modelling and interpreting data in order to investigate and understand real-world phenomena and solve problems in context. Together, mathematics and statistics provide a framework for thinking and a means of communication that is powerful, logical, concise and precise.

The major themes of the Mathematics Methods ATAR course are calculus and statistics. They include, as necessary prerequisites, studies of algebra, functions and their graphs, and probability. They are developed systematically, with increasing levels of sophistication and complexity. Calculus is essential for developing an understanding of the physical world because many of the laws of science are relationships involving rates of change. Statistics is used to describe and analyse phenomena involving uncertainty and variation. For these reasons, this course provides a foundation for further studies in disciplines in which mathematics and statistics have important roles. It is also advantageous for further studies in the health and social sciences. This course is designed for students whose future pathways may involve mathematics and statistics and their applications in a range of disciplines at the tertiary level.

Aims

The Mathematics Methods ATAR course aims to develop students':

- understanding of concepts and techniques drawn from algebra, the study of functions, calculus, probability and statistics
- ability to solve applied problems using concepts and techniques drawn from algebra, functions, calculus, probability and statistics
- reasoning in mathematical and statistical contexts and interpretation of mathematical and statistical information, including ascertaining the reasonableness of solutions to problems
- capacity to communicate in a concise and systematic manner using appropriate mathematical and statistical language
- capacity to choose and use technology appropriately and efficiently.

Organisation

This course is organised into a Year 11 syllabus and a Year 12 syllabus. The cognitive complexity of the syllabus content increases from Year 11 to Year 12.

Structure of the syllabus

The Year 11 syllabus is divided into two units, each of one semester duration, which are typically delivered as a pair. The notional time for each unit is 55 class contact hours.

Organisation of content

Unit 1

Contains the three topics:

- Functions and graphs
- Trigonometric functions
- Counting and probability.

Unit 1 begins with a review of the basic algebraic concepts and techniques required for a successful introduction to the study of functions and calculus. Simple relationships between variable quantities are reviewed, and these are used to introduce the key concepts of a function and its graph. The study of probability and statistics begins in this unit with a review of the fundamentals of probability, and the introduction of the concepts of conditional probability and independence. The study of the trigonometric functions begins with a consideration of the unit circle using degrees and the trigonometry of triangles and its application. Radian measure is introduced, and the graphs of the trigonometric functions are examined and their applications in a wide range of settings are explored.

Unit 2

Contains the three topics:

- Exponential functions
- Arithmetic and geometric sequences and series
- Introduction to differential calculus.

In Unit 2, exponential functions are introduced and their properties and graphs examined. Arithmetic and geometric sequences and their applications are introduced and their recursive definitions applied. Rates and average rates of change are introduced and this is followed by the key concept of the derivative as an ‘instantaneous rate of change’. These concepts are reinforced numerically (by calculating difference quotients), geometrically (as slopes of chords and tangents), and algebraically. This first calculus topic concludes with derivatives of polynomial functions, using simple applications of the derivative to sketch curves, calculate slopes and equations of tangents, determine instantaneous velocities, and solve optimisation problems.

Each unit includes:

- a unit description – a short description of the focus of the unit
- learning outcomes – a set of statements describing the learning expected as a result of studying the unit
- unit content – the content to be taught and learned.

Role of technology

It is assumed that students will be taught this course with an extensive range of technological applications and techniques. If appropriately used, these have the potential to enhance the teaching and learning of the course. However, students also need to continue to develop skills that do not depend on technology. The ability to be able to choose when or when not to use some form of technology and to be able to work flexibly with technology are important skills in this course.

Week/s	Essential Content	ACARA Reference and Elaborations	Reference and Resources	Cross Curricular Priorities / General Capabilities	Assessment
1-2 Term I 2018	Lines and linear relationships.	1.1.1 Determine the coordinates of the mid-point between two points. 1.1.2 Determine an end-point given the other end-point and the mid-point. 1.1.3 Examine examples of direct proportion and linearly related variables. 1.1.4 Recognise features of the graph of $y = mx + c$, including its linear nature, its intercepts and its slope or gradient. 1.1.5 Determine the equation of a straight line given sufficient information; including parallel and perpendicular lines. 1.1.6 Solve linear equations, including those with algebraic fractions and variables on both sides.	Cambridge Senior Mathematical Methods Yr 11 **Ch 1&2**		
3-4 Term I 2018	Quadratic relationships	1.1.7 Examine examples of quadratically related variables 1.1.8 Recognise features of the graphs of $y = x^2$, $y = a(x-b)^2 + c$, and $y = a(x-b)(x-c)$, including their parabolic nature, turning points, axes of symmetry and intercepts 1.1.9 Solve quadratic equations, including the use of quadratic formula and completing the square 1.1.10 Determine the equation of a quadratic given sufficient information 1.1.11 Determine turning points and zeros of quadratics and understand the role of the discriminant 1.1.12 Recognise features of the graph of the general quadratic $y = ax^2 + bx + c$	Cambridge Senior Mathematical Methods Yr 11 **Ch 3**		**TEST 1**
5 Term I 2018	Graphs	1.1.14 Recognise features and determine equations of the graphs of $y = \frac{1}{x}$ and $y = \frac{a}{x-b}$, including their hyperbolic shapes and their asymptotes. 1.1.15 Recognise features of the graphs of $y = x^n$ for $n \in \boldsymbol{N}, n = -1$ and $n = ½$, including shape, and behaviour as $x \to \infty$ and $x \to -\infty$ 1.1.16 Identify the coefficients and the degree of a polynomial 1.1.21 Recognise features and determine equations of the graphs of $x^2 + y^2 = r^2$ and $(x-a)^2 + (y-b)^2 = r^2$, including their circular shapes, their centres and their radii 1.1.22 Recognise features of the graph of $y^2 = x$, including its parabolic shape and its axis of symmetry	Cambridge Senior Mathematical Methods Yr 11 **Ch 4**		
6 Term I 2018	Inverse proportion	1.1.3 Examine examples of direct proportion and linearly related variables. 1.1.13 Examine examples of inverse proportion	Cambridge Senior Mathematical Methods Yr 11 **Ch 5**		**EPW 1**
7 Term I 2018	Functions relations and transformations	1.1.23 Understand the concept of a function as a mapping between sets and as a rule or a formula that defines one variable quantity in terms of another 1.1.24 Use function notation; determine domain and range; recognise independent and dependent variables	Cambridge Senior Mathematical Methods Yr 11 **Ch 6**		
8 Term I 2018	Functions relations and transformations	1.1.25 Understand the concept of the graph of a function 1.1.28 Recognise the distinction between functions and relations and apply the vertical line test 1.1.26 Examine translations and the graphs of $y = f(x) + a$ and $y = f(x-b)$ 1.1.27 Examine dilations and the graphs of $y = cf(x)$ and $y = f(dx)$	Cambridge Senior Mathematical Methods Yr 11 **Ch 6**		
9-10 Term I 2018	Functions relations and transformations	1.1.17 Expand quadratic and cubic polynomials from factors 1.1.18 Recognise features and determine equations of the graphs of $y = x^3$, $y = a(x-b)^3 + c$ and $y = k(x-a)(x-b)(x-c)$, including shape, intercepts and behaviour as $x \to \infty$ and $x \to -\infty$ 1.1.19 Factorise cubic polynomials in cases where a linear factor is easily obtained	Cambridge Senior Mathematical Methods Yr 11 **Ch 7**		

		1.1.20 Solve cubic equations using technology, and algebraically in cases where a linear factor is easily obtained			
11-12 Term II 2018	Probability	1.3.6 Review the concepts and language of outcomes, sample spaces, and events, as sets of outcomes 1.3.7 Use set language and notation for events, including: $\bar{A}$ (or A') for the complement of an event A $A \cap B$ and $A \cup B$ for the intersection and union of events A and B respectively $A \cap B \cap C$ and $A \cup B \cup C$ for the intersection and union of the three events A, B and C respectively recognise mutually exclusive events. 1.3.8 Use everyday occurrences to illustrate set descriptions and representations of events and set operations 1.3.9 Review probability as a measure of 'the likelihood of occurrence' of an event 1.3.10 Review the probability scale: $0 \leq P(A) \leq 1$ for each event A, with $P(A) = 0$ if A is an impossibility and $P(A) = 1$ if A is a certainty 1.3.11 Review the rules: $P(\overline{\mathrm{A}}) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ 1.3.12 Use relative frequencies obtained from data as estimates of probabilities	Cambridge Senior Mathematical Methods Yr 11 **Ch 9**		
13 Term II 2018	Probability	1.3.13 Understand the notion of a conditional probability and recognise and use language that indicates conditionality 1.3.14 Use the notation $P(A\|B)$ and the formula $P(A \cap B) = P(A\|B)P(B)$ 13.15 Understand the notion of independence of an event A from an event B, as defined by $P(A\|B) = P(A)$ 1.3.16 Establish and use the formula $P(A \cap B) = P(A)P(B)$ for independent events A and B, and recognise the symmetry of independence 1.3.17 Use relative frequencies obtained from data as estimates of conditional probabilities and as indications of possible independence of events	Cambridge Senior Mathematical Methods Yr 11 **Ch 9**		**TEST 3**
14 Term II 2018		Revision			
15 – 16 Term II 2018		**SEMESTER 1 EXAMS**			EXAM

Unit 2

Week/s	Essential Content	Elaborations	Reference and Resources	Cross Curricular Priorities / General Capabilities	Assessment
17 Term II 2018	Combinations, Counting and probability	1.3.1 Understand the notion of a combination as a set of r objects taken from a set of n distinct objects 1.3.2 Use the notation $\binom{n}{r}$ and the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ for the number of combinations of r objects taken from a set of n distinct objects 1.3.1 Expand $(x + y)^n$ for small positive integers n 1.3.4 Recognise the numbers $\binom{n}{r}$ as binomial coefficients (as coefficients in the expansion of $(x + y)^n$ 1.3.5 Use Pascal's triangle and its properties	Cambridge Senior Mathematical Methods Yr 11 **Ch 10**		
18 Term II 2018	Circular Functions	1.2.7 Understand the unit circle definition of $\sin\theta$, $\cos\theta$ and $\tan\theta$ and periodicity using radians 1.2.8 Recognise the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ at integer multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$ 1.2.9 Recognise the graphs of $y = \sin x, y = \cos x$, and $y = \tan x$ on extended domains 1.2.10 Examine amplitude changes and the graphs of $y = a\sin x$ and $y = a\cos x$ 1.2.11 Examine period changes and the graphs of $y = \sin bx$, $y = \cos bx$ and $y = \tan\ bx$ 1.2.12 Examine phase changes and the graphs of $y = \sin(x-c)$, $y = \cos(x-c)$ and $y = \tan(x-c)$ 1.2.13 Examine the relationships $\sin\left(x + \frac{\pi}{2}\right) = \cos x$ and $\cos\left(x - \frac{\pi}{2}\right) = \sin x$ 1.2.14 Prove and apply the angle sum and difference identities 1.2.15 Identify contexts suitable for modelling by trigonometric functions and use them to solve practical problems 1.2.16 Solve equations involving trigonometric functions using technology, and algebraically in simple cases	Cambridge Senior Mathematical Methods Yr 11 **Ch 12**		
19 Term II 2018	Trig ratios and applications	1.2.4 Establish and use the cosine and sine rules, including consideration of the ambiguous case and the formula $Area = \frac{1}{2}bc\ sin\ A$ for the area of a triangle	Cambridge Senior Mathematical Methods Yr 11 **Ch 13**		**EPW 2**
20 Term II 2018	Exponential functions Exponential functions	2.1.1 Establish and use the algebraic properties of exponential functions 2.1.2 Recognise the qualitative features of the graph of $y = a^x$ $(a > 0)$, including asymptotes, and of its translations ($y = a^x + b$ and $y = a^{x-c}$) 2.1.3 Identify contexts suitable for modelling by exponential functions and use them to solve practical problems 2.1.4 Establish and use the algebraic properties of exponential functions 2.1.5 recognise the qualitative features of the graph of $y = a^x$ $(a > 0)$, including asymptotes, and of its translations ($y = a^x + b$ and $y = a^{x-c}$) 2.1.6 Identify contexts suitable for modelling by exponential functions and use them to solve practical problems 2.1.7 Solve equations involving exponential functions using technology, and algebraically in simple cases	Cambridge Senior Mathematical Methods Yr 11 **Ch 14**		
21	Arithmetic and geometric	2.2.1 Recognise and use the recursive definition of an arithmetic sequence: $t_{n+1} = t_n + d$	Cambridge Senior Mathematical		

Term II 2018	sequences and series Arithmetic sequences	2.2.2 Develop and use the formula $t_n = t_1 + (n-1)d$ for the general term of an arithmetic sequence and recognise its linear nature 2.2.3 Use arithmetic sequences in contexts involving discrete linear growth or decay, such as simple interest 2.2.4 Establish and use the formula for the sum of the first n terms of an arithmetic sequence	Methods Yr 11 **Ch 15**		
22 Term III 2018	Arithmetic and geometric sequences and series Geometric sequences	2.2.5 Recognise and use the recursive definition of a geometric sequence: $t_{n+1} = t_n \times r$ 2.2.6 Develop and use the formula $t_n = t_1 \times r^{n-1}$ for the general term of a geometric sequence and recognise its exponential nature 2.2.7 Understand the limiting behaviour as $n \to \infty$ of the terms t_n in a geometric sequence and its dependence on the value of the common ratio r 2.2.8 Establish and use the formula $S_n = t_1 \frac{r^n - 1}{r - 1}$ for the sum of the first n terms of a geometric sequence 2.2.9 Use geometric sequences in contexts involving geometric growth or decay, such as compound interest	Cambridge Senior Mathematical Methods Yr 11 **Ch 15**		
23 Term III 2018	Introduction to differential calculus Rates of change	2.3.1 Interpret the difference quotient $\frac{f(x+h)-f(x)}{h}$ as the average rate of change of a function f 2.3.2 Use the Leibniz notation δx and δy for changes or increments in the variables x and y 2.3.3 Use the notation $\frac{\delta y}{\delta x}$ for the difference quotient $\frac{f(x+h)-f(x)}{h}$ where $y = f(x)$ 2.3.4 Interpret the ratios $\frac{f(x+h)-f(x)}{h}$ and $\frac{\delta y}{\delta x}$ as the slope or gradient of a chord or secant of the graph of $y = f(x)$	Cambridge Senior Mathematical Methods Yr 11 **Ch 17**		**TEST 4**
24 Term III 2018	Introduction to differential calculus The concept of the derivative	2.3.5 Examine the behaviour of the difference quotient $\frac{f(x+h)-f(x)}{h}$ as $h \to 0$ as an informal introduction to the concept of a limit 2.3.6 Define the derivative $f'(x)$ as $\lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$ 2.3.7 Use the Leibniz notation for the derivative: $\frac{dy}{dx} = \lim_{\delta x \to 0} \frac{\delta y}{\delta x}$ and the correspondence $\frac{dy}{dx} = f'(x)$ where $y = f(x)$ 2.3.8 Interpret the derivative as the instantaneous rate of change 2.3.9 Interpret the derivative as the slope or gradient of a tangent line of the graph of $y = f(x)$	Cambridge Senior Mathematical Methods Yr 11 **Ch 17**		
25 Term III 2018	Introduction to differential calculus Computation of derivatives Properties of derivatives	2.3.10 Estimate numerically the value of a derivative for simple power functions 2.3.11 Examine examples of variable rates of change of non-linear functions 2.3.12 Establish the formula $\frac{d}{dx}(x^n) = nx^{n-1}$ for non-negative integers n expanding $(x+h)^n$ or by factorising $(x+h)^n - x^n$ 2.3.13 Understand the concept of the derivative as a function 2.3.14 Identify and use linearity properties of the derivative	Cambridge Senior Mathematical Methods Yr 11 **Ch 17**		

		2.3.15 Calculate derivatives of polynomials			
26 Term III 2018	Introduction to differential calculus Applications of derivatives	2.3.16 Determine instantaneous rates of change 2.3.17 Determine the slope of a tangent and the equation of the tangent	Cambridge Senior Mathematical Methods Yr 11 **Ch 17**		
27 Term III 2018	Introduction to differential calculus Applications of derivatives Anti-derivatives	2.3.22 Calculate anti-derivatives of polynomial functions 2.3.20 Sketch curves associated with simple polynomials, determine stationary points, and local and global maxima and minima, and examine behaviour as $x \to \infty$ and $x \to -\infty$ 2.3.21 Solve optimisation problems arising in a variety of contexts involving polynomials on finite interval domains	Cambridge Senior Mathematical Methods Yr 11 **Ch 18**		
28 Term III 2018	Introduction to differential calculus Applications of derivatives	2.3.18 Construct and interpret position-time graphs with velocity as the slope of the tangent 2.3.19 Recognise velocity as the first derivative of displacement with respect to time			
29					**EPW 3**
30-32 Term III 2018		Revision			**TEST 5**
33 to 34 Term IV 2018		EXAMS			

Icon Key

Habits of Mind

Persisting	Manage Impulsivity	Understanding and empathy	Thinking flexibly	Think about thinking	Strive for Accuracy	Posing Questions	Apply past knowledge
Clarity and precision	Use all senses	Imagining and Creating	Wonderment and awe	Responsible risk taking	Find humour	Think interdependently	Continuous learning

General Capabilities

Literacy	Numeracy	Critical and Creative thinking	ICT	Social Capability	Intercultural understanding	Ethical Understanding

Cross Curricular Priorities

Aboriginal and Torres Strait Islander Cultures	Asia and Australia's engagement	Sustainability

Year 11 Mathematics Methods

Assessment Outline

Test One	8%	Term 1 Week 4
Test Two	8%	Term 1 Week 10
Test Three	8%	Term 2 Week 2
Test Four	8%	Term 3 Week 3
Test Five	8%	Term 3 Week 10
TOTAL TESTS	**40%**	
Investigation One	6%	Term 1 Week 6
Investigation Two	7%	Term 2 Week 8
Investigation Three	7%	Term 3 Week 9
TOTAL INVESTIGATIONS	**20%**	
Examination One	16%	Term 2 Week 4/5
Examination Two	24%	Term 4 Week 3/4
TOTAL EXAMINATIONS	**40%**	

Achievement Standards Units 1 and 2

Concepts and Techniques

A	B	C	D	E
• demonstrates knowledge of concepts of functions, calculus and statistics in routine and non-routine problems in a variety of contexts	• demonstrates knowledge of concepts of functions, calculus and statistics in routine and non-routine problems	• demonstrates knowledge of concepts of functions, calculus and statistics that apply to routine problems	• demonstrates knowledge of concepts of simple functions, calculus and statistics	• demonstrates limited familiarity with concepts of simple functions, calculus and statistics
• selects and applies techniques in functions, calculus and statistics to solve routine and non-routine problems in a variety of contexts	• selects and applies techniques in functions, calculus and statistics to solve routine and non-routine problems	• selects and applies techniques in functions, calculus and statistics to solve routine problems	• uses simple techniques in functions, calculus and statistics in routine problems	• uses simple techniques in a structured context
• develops, selects and applies mathematical and statistical models in routine and non-routine problems in a variety of contexts	• selects and applies mathematical and statistical models in routine and non-routine problems	• applies mathematical and statistical models in routine problems	• demonstrates familiarity mathematical and statistical models	• demonstrates limited familiarity with mathematical or statistical models
• uses digital technologies effectively to graph, display and organise mathematical and statistical information and to solve a range of routine and non-routine problems in a variety of contexts	• uses digital technologies appropriately to graph, display and organise mathematical and statistical information and to solve a range of routine and non-routine problems	• uses digital technologies to graph, display and organise mathematical and statistical information to solve routine problems	• uses digital technologies to display some mathematical and statistical information in routine problems	• uses digital technologies for arithmetic calculations and to display limited mathematical and statistical information

Reasoning and Communication

A	B	C	D	E
• represents functions, calculus and statistics in numerical, graphical and symbolic form in routine and non-routine problems in a variety of contexts	• represents functions, calculus and statistics in numerical, graphical and symbolic form in routine and non-routine problems	• represents functions, calculus and statistics in numerical, graphical and symbolic form in routine problems	• represents simple functions and distributions in numerical, graphical or symbolic form in routine problems	• represents limited mathematical or statistical information in a structured context
• communicates mathematical and statistical judgments and arguments, which are succinct and reasoned, using appropriate language	• communicates mathematical and statistical judgments and arguments, which are clear and reasoned, using appropriate language	• communicates mathematical and statistical arguments using appropriate language	• communicates simple mathematical and statistical information using appropriate language	• communicates simple mathematical and statistical information
• interprets the solutions to routine and non-routine problems in a variety of contexts	• interprets the solutions to routine and non-routine problems	• interprets the solutions to routine problems	• describes solutions to routine problems	• identifies solutions to routine problems
• explains the reasonableness of the results and solutions to routine and non-routine problems in a variety of contexts	• explains the reasonableness of the results and solutions to routine and non-routine problems	• describes the reasonableness of results and solutions to routine problems	• describes the appropriateness of the result of calculations	• describes with limited familiarity the appropriateness of the results of calculations
• identifies and explains the validity and limitations of models used when developing solutions to routine and non-routine problems	• identifies and explains the limitations of models used when developing solutions to routine problems	• identifies the limitations of models used when developing solutions to routine problems	• identifies the limitations of simple models used	• identifies simple models